

TST for IPhO 2025

Team Italy

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1 Models for Material Response Functions [100 points]

In vacuum, a point particle of charge q , moving with velocity $\vec{v}$ and subject to external fields $\vec{E}_{\text{ext}}$ and $\vec{B}_{\text{ext}}$, experiences the force

$$\vec{F} = q \left(\vec{E}_{\text{ext}} + \vec{v} \times \vec{B}_{\text{ext}} \right).$$

In matter, however, things are more complicated. In fact, the action of an external field on the system causes the creation of induced charges and currents, whose field modifies the value of the total field. More precisely, when the external field $\vec{E}_{\text{ext}}$ is applied, the so-called *polarization charges* that form generate the induced field $\vec{E}_{\text{ind}}$, so the total electric field in the system is

$$\vec{E}_{\text{tot}} = \vec{E}_{\text{ext}} + \vec{E}_{\text{ind}}.$$

The same happens in the magnetic case, although instead of polarization charges there are *magnetization currents*.

As an example, consider the case of an infinite slab of thickness l , permeated by an external electric field generated by an infinite charged plane, as shown in Fig. 1.

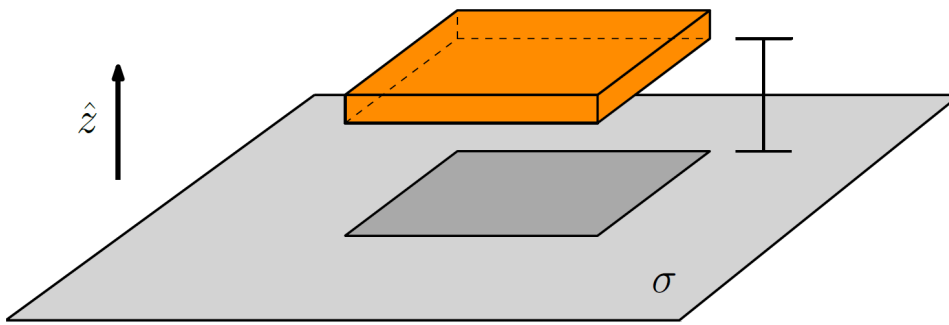


Figura 1: Configuration of the dielectric slab, which, although not shown in the figure, is infinite.

The total electric field inside the slab is equal to the sum of the external field and the induced field generated by the polarization charges formed on the slab. Explicitly:

$$\vec{E}_{\text{tot}} = \vec{E}_{\text{ext}} + \vec{E}_{\text{ind}} = \frac{\sigma}{2\varepsilon_0} \hat{z} + \vec{E}_{\text{ind}} = \frac{\sigma}{2\varepsilon_0\varepsilon_r} \hat{z},$$

where ε_r is the so-called *relative dielectric constant*, whose value depends on the properties of the material composing the slab. The same effect occurs for a point charge placed in a

dielectric: the external field is that generated by the charge in vacuum,

$$\vec{E}_{\text{ext}} = \frac{q}{4\pi\epsilon_0 r^2} \hat{r},$$

while the total field is

$$\vec{E}_{\text{tot}} = \frac{q}{4\pi\epsilon_0\epsilon_r r^2} \hat{r}.$$

In the magnetic case, in contrast, one must introduce the relative permeability μ_r , whose value distinguishes the regimes of diamagnetism ($0 < \mu_r < 1$) and paramagnetism ($\mu_r \geq 1$).

The goal of this problem is to construct two models that describe how ϵ_r and its magnetic counterpart μ_r depend on the microscopic quantities of matter in the limit of *low atomic densities*. For simplicity, we restrict to non-polar materials, meaning those in which the dipole moment of the microscopic constituents—atoms and molecules—is zero when external fields are not present.

Electric case

An atom consists of a point nucleus of charge Ze , where Z is the atomic number and e is the elementary charge. We model the electronic cloud around the nucleus as a rigid homogeneous sphere of total charge $-Ze$ and radius a , centered on the nucleus.

1. (9 pt.) When a sufficiently small and homogeneous external electric field E_{ext} is applied, the center of the charged sphere moves until it reaches a new equilibrium position. Compute the distance d between the center of the sphere and the nucleus as a function of E_{ext} . Also explain what would happen if the electric field were arbitrarily large, and provide a physical interpretation of the phenomenon.
2. (5 pt.) What is the magnitude of the electric dipole moment associated with the equilibrium configuration found above?

Let us consider again the slab in Fig. 1.

3. (7 pt.) Using the atomic model described earlier, explain why polarization charges form only on the surfaces of the slab and not within its interior.
4. (10 pt.) If the slab contains n atoms per unit volume, what is the induced surface charge σ_{ind} on the slab's surfaces? Express it in terms of the surface charge density σ of the plane and the parameters introduced earlier.
5. (9 pt.) Find an expression for ϵ_r that depends only on a and n .

Magnetic case

Consider a thin homogeneous ring of total charge Q and mass M , rotating around its axis.

6. (8 pt.) Show that the angular momentum $\vec{L}$ and the magnetic dipole moment $\vec{m}$ are related by

$$\vec{m} = g\vec{L},$$

where g is the so-called *gyromagnetic ratio*. Determine the value of g and explain why the same relation holds for any uniformly charged solid of revolution with homogeneous mass density. *The magnetic dipole moment of a planar loop carrying a current i is $\vec{m} = i\vec{S}$, where $\vec{S}$ is the area vector, whose magnitude equals the area enclosed by the loop, with direction perpendicular to the plane of the loop and orientation given by the right-hand rule.*

From now until the end of the problem, we consider a monoatomic material with even Z , where each electron can be modeled as a sphere of uniform charge distribution and radius a , rotating around an axis passing through the nucleus, with angular velocity $\vec{\omega}$. For every electron, there exists another one rotating around the same axis but in the opposite direction.

7. (6 pt.) Let m_e be the mass of the electron. Compute the magnetic dipole moment $\vec{m}$ associated with a single electron, and explain why the total magnetic dipole moment associated with the atom is zero in the absence of external fields.
8. (13 pt.) When an external magnetic field is applied, the situation changes. A body with a non-zero magnetic dipole moment experiences a torque when placed in a region permeated by an external magnetic field. Suppose $\vec{B}_{\text{ext}}$ is homogeneous. If the rotation axis of the electronic sphere is not aligned with $\vec{B}_{\text{ext}}$, it begins to precess around the direction of $\vec{B}_{\text{ext}}$ like a spinning top. Compute the angular precession frequency ω_L .

The precession phenomenon, averaged over a period, induces a current in the plane orthogonal to the axis of precession, i.e., along the direction of $\vec{B}_{\text{ext}}$. Suppose we can model this current as if the entire charge moves along a circular ring of radius $a/2$.

9. (13 pt.) Compute the dipole moment $\vec{m}_{\text{ind}}$ associated with the induced current just described.

Consider now an infinite solenoid with ν turns per unit length, extending along the z -axis and carrying a current i . Inside the solenoid is inserted an infinite, solid, homogeneous cylinder composed of a material susceptible to external magnetic fields. The cylinder and the solenoid share the same axis. In this case, analogously to the electric case,

$$\vec{B}_{\text{ext}} = \mu_0 \nu i \hat{z}$$

and

$$\vec{B}_{\text{tot}} = \vec{B}_{\text{ext}} + \vec{B}_{\text{ind}} = \mu_0 \nu i \hat{z} + \vec{B}_{\text{ind}} = \mu_0 \mu_r \nu i \hat{z}.$$

As in the electric case, we restrict to the situation where the total magnetic moment of each atom, with the solenoid turned off, is zero.

10. (10 pt.) Let n be the atomic number density. Show that the effect of the induced magnetic dipole moments is equivalent to that of a solenoid carrying a current surrounding the cylinder. Determine the induced magnetic field inside the cylinder as a function of the external magnetic field and the parameters given above.
11. (10 pt.) Find an expression for μ_r that depends only on n , the atomic characteristics, and the fundamental constants. Is this a case of diamagnetism or paramagnetism?

2 Classical Model of the Expansion of the Universe [100 points]

From General Relativity one obtains the so-called *Friedmann Equations*, which describe the expansion of the universe. In this problem we will obtain similar results using simpler tools.

Let us model the universe as a sphere with density $\rho(t)$, time-dependent but uniform inside the sphere. Let $R(t)$ be the radius of the sphere at time t , and let $\dot{R}(t)$ and $\ddot{R}(t)$ be its first and second time derivatives. The expansion or contraction of the sphere occurs

uniformly, as if the sphere underwent a similarity transformation. In other words, a point that at time t_0 is at a distance $r(t_0)$ from the center of the sphere will, at time t_1 , be at the distance

$$r(t_1) = r(t_0) \frac{R(t_1)}{R(t_0)}.$$

1. (2 pt.) Assuming that the total mass of the universe M is conserved, determine the value of $\rho(t)$ as a function of $R(t)$.
2. (8 pt.) Compute the total kinetic energy of the sphere, as a function of M and $\dot{R}(t)$.

In problems of this kind, since one considers the entire universe, it is important to specify the domain of the functions and the boundary conditions. To compute the gravitational energy it is necessary to choose a reference point for the potential, usually a zero. From now until the end of the problem, for ALL requests (3→8) you must consider 3 different possibilities for the choice of the potential:

- A. $V = 0$ at $r = 0$,
- B. $V = 0$ at $r = R(t)$,
- C. $V = 0$ for $r \rightarrow \infty$.

Thus, each of the following questions requires THREE different answers, each corresponding to one of the three cases described above. In the first two cases, space is considered finite, limited to the sphere of radius $R(t)$, while in the last case the vacuum surrounding the sphere and extending to infinity must also be considered.

3. (24 pt.) Compute the gravitational field at every point in space and the corresponding potential, indicating clearly the dependence on the radius of the universe $R(t)$.
4. (27 pt.) Compute the total gravitational potential energy of the universe.
5. (9 pt.) Express the value of the quantity

$$H^2 \equiv \frac{\dot{R}^2(t)}{R^2(t)}$$

as a function of $R(t)$, M , and E_0 , where E_0 is the initial mechanical energy of the universe. Are there constraints on the sign of E_0 in the three cases?

6. (9 pt.) Express the value of the *deceleration parameter* $q \equiv -\frac{\ddot{R}(t)R(t)}{\dot{R}^2(t)}$ as a function of $R(t)$. Also discuss the sign of q in the three cases.
7. (15 pt.) Discuss qualitatively the motion of the sphere in the three cases. In the cases where the radius of the sphere undergoes periodic motion, explicitly find the period as a function of M and E_0 .
8. (6 pt.) Which cases among A, B, and C are physically equivalent, up to a redefinition of the units of measurement? Two situations are said to be *physically equivalent* if, for each of the seven fundamental units, the ratio between the corresponding fundamental quantities u_1 and u_2 measured in the two situations is the same in all 7 cases.

3 Energy Levels [100 points]

Part 1: Three-level System

Under suitable assumptions, every component of a thermodynamic system, such as an atom or a molecule, obeys the Boltzmann distribution. The probability p_i that, at a given temperature T , one of these elements is found in a particular state i with energy E_i is

$$p_i \propto e^{-E_i/k_B T},$$

where k_B is Boltzmann's constant.

Consider a system composed of $N \gg 1$ identical non-interacting atoms, in thermal equilibrium with a heat bath at temperature T . Each atom can be in one of the three energy levels shown in Fig. 2, corresponding respectively to the energies $E_1 = \varepsilon$, $E_2 = 3\varepsilon$, and $E_3 = 10\varepsilon$, with $\varepsilon > 0$.



Figura 2: Three-level system.

1. (7 pt.) Let N_1 , N_2 , and N_3 be the average numbers of atoms in the respective energy states. Find N_1 , N_2 , and N_3 as functions of N , ε , and T .
2. (16 pt.) Find the average energy of a single atom as a function of N , ε , and T , and sketch a qualitative plot as a function of the temperature over the domain $0 \leq T < \infty$. In the plot, specify the coordinates of notable points, indicate any asymptotes, and describe qualitatively the behavior of the slope (positive, negative, or zero) throughout the domain.
3. (14 pt.) In systems at thermodynamic equilibrium one can define a heat capacity,

$$C(T) = \frac{dE_{\text{tot}}}{dT}.$$

Compute the heat capacity of the three-level system and evaluate its asymptotic behavior for $T \rightarrow 0$ and $T \rightarrow \infty$, i.e., the functions that best approximate the behavior of $C(T)$ in the two limits.

4. (10 pt.) If the temperature of the system drops below a critical value T^* , only the first two energy levels are populated. Estimate T^* as a function of N and ε .

Part 2: Einstein Coefficients

In the previous section the atomic system was isolated from external perturbations. The goal of this section is to study what happens in a two-level system when electromagnetic radiation is considered, as it can induce transitions between energy levels.

Consider a two-level system with energies E_1 and $E_2 > E_1$, different from those in the previous section.

5. (3 pt.) A photon is absorbed by an atom in the lower energy state, giving it the minimum energy needed to be promoted to the upper level. What is the frequency ν of the photon?

In addition to promotion by photon absorption, two other phenomena are possible: an atom in the upper energy level may emit a photon and decay to the lower level via *stimulated emission* or *spontaneous emission*. The first is induced by an external photon, while the second is a random process requiring no mediator.

Let $N_1(t)$ and $N_2(t)$ be the numbers of atoms in the respective energy levels at time t , and let

$$\rho(\nu) = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$$

be the spectral energy density of electromagnetic radiation, corresponding to the same frequency ν computed in point 5. The following facts hold:

- The number of atoms per second that decay from $2 \rightarrow 1$ via spontaneous emission is $A_{21}N_2(t)$, where A_{21} is a positive constant called the *spontaneous emission coefficient*.
 - The number of atoms per second that are promoted from $1 \rightarrow 2$ is $B_{12}N_1(t)\rho(\nu)$, where B_{12} is a positive constant called the *absorption coefficient*.
 - The number of atoms per second that decay from $2 \rightarrow 1$ via stimulated emission is $B_{21}N_2(t)\rho(\nu)$, where B_{21} is a positive constant called the *stimulated emission coefficient*.
6. (15 pt.) Write the equations for the rates of change $\frac{dN_1}{dt}$ and $\frac{dN_2}{dt}$ as functions of $N_1(t)$, $N_2(t)$, A_{21} , B_{12} , B_{21} , and $\rho(\nu)$.
7. (7 pt.) Find the value of the ratio N_2/N_1 at equilibrium, as a function of A_{21} , B_{12} , B_{21} , and $\rho(\nu)$.

The coefficients A_{21} , B_{12} , and B_{21} cannot take arbitrary values, but are related to each other.

8. (14 pt.) Find the value of the ratio A_{21}/B_{12} as a function of the frequency ν and the fundamental constants.
9. (14 pt.) Find the value of the ratio B_{12}/B_{21} as a function of the frequency ν and the fundamental constants.